

Creating a home page and 2 applications to be opened from the home page

Step 1: Open Workspace and Set Execution Policy

Open VS Code, select File > Open Folder, and create or select your workspace folder. Open your terminal (Ctrl + ~) and run this command to allow script execution for your current session:

PowerShell

Set-ExecutionPolicy -ExecutionPolicy RemoteSigned -Scope Process

Step 2: Create and Activate the Virtual Environment

Run the following commands to create a virtual environment named venv and activate it:

PowerShell

python -m venv venv

.\venv\Scripts\Activate

(You will notice a (venv) prefix appear on your terminal line, confirming you are safely isolated).

Step 3: Install Django and Start the Project

Install Django via pip and initialize myproject directly at your workspace root:

PowerShell

pip install django

django-admin startproject collegeproject .

Once you create the main project, you need to create the home application from where you can track the remaining applications

python manage.py startapp collegeapp

Open collegeproject/settings.py and add

```
INSTALLED_APPS = [  
    'django.contrib.admin',
```

```
'django.contrib.auth',
'django.contrib.contenttypes',
'django.contrib.sessions',
'django.contrib.messages',
'django.contrib.staticfiles',

'collegeapp',
]
```

Then execute this in powershell

python manage.py startapp studentapp

python manage.py startapp facultyapp

Open collegeapp/views.py

```
from django.shortcuts import render

# We will import these models after creating the student
# and faculty applications.
from studentapp.models import Student
from facultyapp.models import Faculty

def home_view(request):

    total_students = Student.objects.count()
    total_faculty = Faculty.objects.count()

    context = {
        'total_students': total_students,
        'total_faculty': total_faculty,
    }

    return render(request, 'dashboardapp/home.html', context)
```

Open collegeapp/templates/collegeapp/home.html

```
<!DOCTYPE html>
<html lang="en">

<head>
    <meta charset="UTF-8">
    <title>College Management System</title>
</head>
```

```
<body>

    <h1>Welcome to College Management System</h1>

    <p>This is the main home page.</p>

    <hr>

    <h2>System Overview</h2>

    <ul>
        <li>
            <strong>Total Students:</strong>
            {{ total_students }}
        </li>

        <li>
            <strong>Total Faculty:</strong>
            {{ total_faculty }}
        </li>
    </ul>

    <hr>

    <h2>Applications</h2>

    <ul>
        <li>
            <a href="/student/">
                Student Application
            </a>
        </li>

        <li>
            <a href="/faculty/">
                Faculty Application
            </a>
        </li>
    </ul>

    <hr>

    <h2>View Records</h2>

    <ul>
```

```

        <li>
            <a href="/student/list/">
                View All Students
            </a>
        </li>

        <li>
            <a href="/faculty/list/">
                View All Faculty
            </a>
        </li>
    </ul>

</body>

</html>
////////////////////////////////////////////////////////////////
<!DOCTYPE html>
<html>
<head>
    <title>College Management</title>
</head>
<body>

<h1>College Management System</h1>

<h3>System Overview</h3>
<p>Total Students: {{ total_students }}</p>
<p>Total Faculty: {{ total_faculty }}</p>

<h3>Applications</h3>
<a href="/student/">Student Application</a><br>
<a href="/faculty/">Faculty Application</a><br><br>

<h3>View Records</h3>
<a href="/student/list/">View Students</a><br>
<a href="/faculty/list/">View Faculty</a>

</body>
</html>

```

```

////////////////////////////////////////////////////////////////

```

Open collegeapp/urls.py

```

from django.urls import path
from . import views

```

```

urlpatterns = [
    path("", views.home_view, name='home'),
]

```

Open studentapp/models.py

```
from django.db import models

class Student(models.Model):

    studID = models.CharField(max_length=20, unique=True)
    studname = models.CharField(max_length=100)
    marks = models.FloatField()

    def __str__(self):
        return self.studID
```

Open studentapp/views.py

```
from django.shortcuts import render, redirect
from .models import Student
```

```
def student_application(request):

    if request.method == 'POST':

        studID = request.POST.get('studID')
        studname = request.POST.get('studname')
        marks = request.POST.get('marks')

        if studID and studname and marks:

            Student.objects.create(
                studID=studID,
                studname=studname,
                marks=marks
            )

            return redirect('student_application')

    students = Student.objects.all()

    return render(
        request,
        'studentapp/student.html',
        {'students': students}
    )
```

```
def student_list(request):

    students = Student.objects.all()

    return render(
        request,
        'studentapp/students.html',
        {'students': students}
    )
```

Create studentapp/templates/studentapp/student.html

```
<!DOCTYPE html>
<html lang="en">

<head>
    <meta charset="UTF-8">
    <title>Student Application</title>
</head>

<body>

    <h2>Student Application</h2>

    <form method="POST">

        {% csrf_token %}

        <label>Student ID:</label>
        <input
            type="text"
            name="studID"
            placeholder="Enter Student ID"
            required
        >

        <br><br>

        <label>Student Name:</label>
        <input
            type="text"
            name="studname"
            placeholder="Enter Student Name"
            required
```

```
>

<br><br>

<label>Marks:</label>
<input
  type="number"
  name="marks"
  placeholder="Enter Marks"
  required
>

<br><br>

<button type="submit">
  Save Student
</button>

</form>

<hr>

<h3>Student Records</h3>

<ul>

  {% for student in students %}

  <li>
    <strong>{{ student.studID }}</strong>
    -
    {{ student.studname }}
    -
    {{ student.marks }}
  </li>

  {% empty %}

  <li>No students added yet.</li>

  {% endfor %}

</ul>

<br>
```

```

        <a href="/">
            Back to Home
        </a>

</body>

</html>
////////////////////////////////////////////////////////////////
<!DOCTYPE html>
<html>
<head>
    <title>Student Application</title>
</head>
<body>

<h2>Student Application</h2>

<form method="POST">
    {% csrf_token %}

    Student ID:
    <input type="text" name="studID" required><br><br>

    Student Name:
    <input type="text" name="studname" required><br><br>

    Marks:
    <input type="number" name="marks" required><br><br>

    <button type="submit">Save Student</button>
</form>

<br>
<a href="/">Back to Home</a>

</body>
</html>
////////////////////////////////////////////////////////////////

```

Create studentapp/urls.py

```

from django.urls import path
from . import views

```

```

urlpatterns = [

```

```

    path(
        "",
        views.student_application,
        name='student_application'
    )

```



```
    ),  
  
    path(  
        'list/',  
        views.student_list,  
        name='student_list'  
    ),  
  
]
```

Open facultyapp/models.py

```
from django.db import models  
  
class Faculty(models.Model):  
  
    facultyID = models.CharField(max_length=20, unique=True)  
    name = models.CharField(max_length=100)  
    designation = models.CharField(max_length=100)  
  
    def __str__(self):  
        return self.facultyID
```

Open facultyapp/views.py

```
from django.shortcuts import render, redirect  
from .models import Faculty  
  
def faculty_application(request):  
  
    if request.method == 'POST':  
  
        facultyID = request.POST.get('facultyID')  
        name = request.POST.get('name')  
        designation = request.POST.get('designation')  
  
        if facultyID and name and designation:  
  
            Faculty.objects.create(  
                facultyID=facultyID,  
                name=name,  
                designation=designation  
            )
```

```

        return redirect('faculty_application')

    faculties = Faculty.objects.all()

    return render(
        request,
        'facultyapp/faculty.html',
        {'faculties': faculties}
    )

def faculty_list(request):

    faculties = Faculty.objects.all()

    return render(
        request,
        'facultyapp/faculties.html',
        {'faculties': faculties}
    )

```

Create facultyapp/templates/facultyapp/faculty.html

```

<!DOCTYPE html>
<html lang="en">

<head>
    <meta charset="UTF-8">
    <title>Faculty Application</title>
</head>

<body>

    <h2>Faculty Application</h2>

    <form method="POST">

        {% csrf_token %}

        <label>Faculty ID:</label>
        <input
            type="text"
            name="facultyID"
            placeholder="Enter Faculty ID"
            required
        >
    </form>

```

```
<br><br>
```

```
<label>Name:</label>
```

```
<input
  type="text"
  name="name"
  placeholder="Enter Faculty Name"
  required
>
```

```
<br><br>
```

```
<label>Designation:</label>
```

```
<input
  type="text"
  name="designation"
  placeholder="Enter Designation"
  required
>
```

```
<br><br>
```

```
<button type="submit">
  Save Faculty
</button>
```

```
</form>
```

```
<hr>
```

```
<h3>Faculty Records</h3>
```

```
<ul>
```

```
{% for faculty in faculties %}
```

```
<li>
  <strong>{{ faculty.facultyID }}</strong>
  -
  {{ faculty.name }}
  -
  {{ faculty.designation }}
</li>
```

```
{% empty %}
```

```

        <li>No faculty added yet.</li>

    {% endfor %}

</ul>

<br>

<a href="/">
    Back to Home
</a>

</body>

</html>
////////////////////////////////////
<!DOCTYPE html>
<html>
<head>
    <title>Faculty Application</title>
</head>
<body>

<h2>Faculty Application</h2>

<form method="POST">
    {% csrf_token %}

    Faculty ID:
    <input type="text" name="facultyID" required><br><br>

    Name:
    <input type="text" name="name" required><br><br>

    Designation:
    <input type="text" name="designation" required><br><br>

    <button type="submit">Save Faculty</button>
</form>

<br>
<a href="/">Back to Home</a>

</body>
</html>
////////////////////////////////////

```

Create facultyapp/urls.py

```

from django.urls import path
from . import views

```

```
urlpatterns = [

    path(
        "",
        views.faculty_application,
        name='faculty_application'
    ),

    path(
        'list/',
        views.faculty_list,
        name='faculty_list'
    ),

]
```

Create facultyapp/templates/facultyapp/faculties.html

```
<!DOCTYPE html>
<html lang="en">

<head>
    <meta charset="UTF-8">
    <title>Faculty Records</title>
</head>

<body>

    <h2>All Faculty Records</h2>

    <table border="1">

        <tr>
            <th>Faculty ID</th>
            <th>Name</th>
            <th>Designation</th>
        </tr>

        {% for faculty in faculties %}

        <tr>
            <td>{{ faculty.facultyID }}</td>
            <td>{{ faculty.name }}</td>
            <td>{{ faculty.designation }}</td>
        </tr>
```

[illegible]

Create studentapp/templates/studentapp/students.html

```
<!DOCTYPE html>
<html lang="en">

<head>
  <meta charset="UTF-8">
  <title>Student Records</title>
</head>

<body>

  <h2>All Student Records</h2>

  <table border="1">

    <tr>
      <th>Student ID</th>
      <th>Student Name</th>
      <th>Marks</th>
    </tr>

    {% for student in students %}

      <tr>
        <td>{{ student.studID }}</td>
        <td>{{ student.studname }}</td>
        <td>{{ student.marks }}</td>
      </tr>

      {% empty %}

      <tr>
        <td colspan="3">
          No student records found.
        </td>
      </tr>

    {% endfor %}

  </table>

  <br>

  <a href="/">
    Back to Home
  </a>
```

```
</body>
```

```
</html>
```

```
////////////////////////////////////
```

```
<!DOCTYPE html>
```

```
<html>
```

```
<body>
```

```
<h2>Student Records</h2>
```

```
<table border="1">
```

```
<tr>
```

```
    <th>ID</th>
```

```
    <th>Name</th>
```

```
    <th>Marks</th>
```

```
</tr>
```

```
{% for student in students %}
```

```
<tr>
```

```
    <td>{{ student.studID }}</td>
```

```
    <td>{{ student.studname }}</td>
```

```
    <td>{{ student.marks }}</td>
```

```
</tr>
```

```
{% endfor %}
```

```
</table>
```

```
<a href="/">Home</a>
```

```
</body>
```

```
</html>
```

```
////////////////////////////////////
```

Add 'studentapp', 'facultyapp', to collegeproject/settings.py

Execute this in powershell

```
python manage.py makemigrations
```

```
python manage.py migrate
```

```
python manage.py runserver
```